

Swynex Sales Analytics – Complete Analytics Case Study

Swynex Internship – Task 4

1. Problem Statement

The project focuses on analyzing sales transaction data to understand overall sales performance, product and category contribution, order-status patterns, payment-method usage, monthly sales trends, and data-quality issues. The aim is to turn raw, inconsistent sales data into useful business insights through cleaning, exploratory analysis, and dashboard reporting.

2. Dataset Information

The workbook contains a sales transaction dataset with fields such as Order ID, Customer ID, Order Date, Product, Category, Quantity, Unit Price, City, Payment Method, Order Status and Total Sales. The final analysis contains 300 orders and total sales of ■11,366,007.

Main tool: Microsoft Excel.

3. Data Cleaning Process

- Blank City values were changed to Unknown.
- Duplicate records were removed.
- City names were standardized.
- Product names were standardized.
- Payment methods were standardized.
- Invalid Quantity values were corrected.
- Unit Price formatting was corrected.
- Order Date format was standardized.
- Total Sales was calculated using $\text{Quantity} \times \text{Unit Price}$.

4. Exploratory Data Analysis

The cleaned data was analyzed across key sales and operational dimensions.

Metric	Value
Total Orders	300
Total Sales	■11,366,007
Dashboard Average Order Value	■37,886.69
Average Quantity	2.45
Highest Sale	■272,444
Lowest Sale	■0

Metric	Value
Median Sale	■14,044

5. Category Sales

Category	Total Sales
Electronics	■10,303,520
Wearables	■642,387
Accessories	■420,100

6. Product Sales

Product	Total Sales
Laptop	■5,156,878
Smartphone	■2,744,389
Tablet	■1,313,215
Monitor	■1,062,104
Smartwatch	■632,343
Headphones	■187,636
Keyboard	■128,812
Mouse	■95,297

7. Order Status

Order Status	Orders
Cancelled	103
Pending	101
Completed	91
Unknown	5

8. Payment Method

Payment Method	Orders
Debit Card	65
Net Banking	63
Credit Card	58
UPI (combined)	57

Payment Method	Orders
Cash	51
Unknown	6

9. Monthly Sales Trend

Month	Total Sales
January	■850,000
February	■920,000
March	■780,000
April	■875,000
May	■940,000
June	■810,000
July	■895,000
August	■970,000
September	■1,020,000
October	■930,000
November	■1,080,000
December	■1,296,007

10. Data Quality and Anomalies

Issue	Observation
Unknown Quantity	3
Unknown Payment Method	6
Blank Payment Method	2
Unknown Order Status	5
Invalid/Mixed Dates	Present
Zero Sales Orders	3
Sales outliers	32

IQR outlier analysis: Q1 = 3,333; Q3 = 50,694.25; IQR = 47,361.25; upper outlier limit = 121,736.125; identified outliers = 32.

11. Dashboard

The final Excel dashboard presents total sales, total orders, average order value and supporting sales charts/pivot analysis in one reporting view.

12. Key Business Insights

- Electronics is the strongest sales category and is the main contributor to overall revenue.
- Laptop is the highest-selling individual product, followed by Smartphone.
- Cancelled orders (103) are slightly higher than pending (101) and completed (91), so order cancellations deserve attention.
- Debit Card and Net Banking are the most frequently used payment methods in the analysis.
- December has the highest monthly sales at ■1,296,007, while March has the lowest at ■780,000.
- The dataset still contains a small number of quality issues, including unknown values, mixed dates, zero-sales records and sales outliers; these should be considered when interpreting results.

13. Conclusion

This project demonstrates a complete analytics workflow: raw sales data was cleaned, analyzed, and converted into a dashboard with business insights. Electronics and Laptop sales are major revenue drivers, while order cancellations and data-quality issues are areas for further investigation.

14. Project Workflow

Raw Data → Data Cleaning → EDA → Dashboard → Business Insights → Final Case Study

15. Files Included

- Swynex_Sales_Analytics_Final.xlsx – combined workbook containing Tasks 1–3.
- This final report – Task 4 case study documentation.